

GLMs for Applied Research

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Session 1: Why go beyond linear regression?

Topics we will cover in this session

- Recap and limits of linear regression
- The GLM framework
- Exponential family
- GLMs estimation and model fit

The linear regression model, briefly

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon, \text{ with } \varepsilon \sim N(0, \sigma^2)$$

- Response, Y , treated as continuous, ranging $(-\infty, +\infty)$
- Errors/residuals, ε , assumed normally distributed
- Mean of Y assumed to change linearly with predictors

Four assumptions we usually take for granted

1. Linear relationship between the dependent and the predictors.
2. Multicollinearity, no strong correlation between predictors.
3. Residual values are normally distributed
4. Homoscedasticity assumes that the variance of the residual is similar across the values of each predictor.

Applied research rarely hands us this

Real outcomes researchers actually collect:

- Yes/no, success/failure
- Counts (hospital visits, relapses, citations)
- Proportions or rates
- Highly skewed positive values (cost, wait time)

None of these naturally behave like an unbounded, constant-variance normal variable.

Why Generalized Linear Models?

Outcome type	Example	Problem with OLS
Binary	Disease yes/no	Predicts P outside [0,1]
Count	Hospital admissions	Predicts negative counts; variance $\neq$ mean
Proportion	% defective items	Bounded at 0 and 1
Positive continuous	Survival time	Skewed; must be > 0

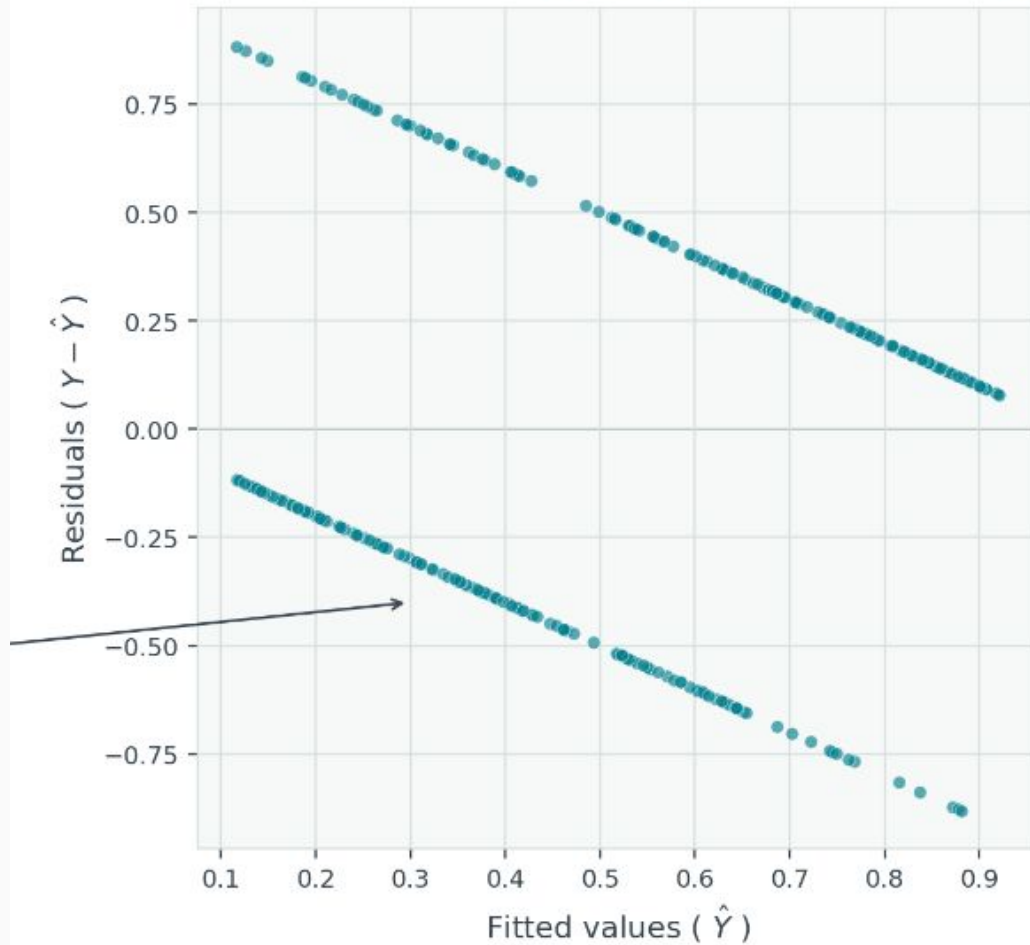
The solution: Generalized Linear Models (GLMs) — a unified framework that extends linear regression to handle non-normal response distributions.

Why this matters: consequences of forcing OLS

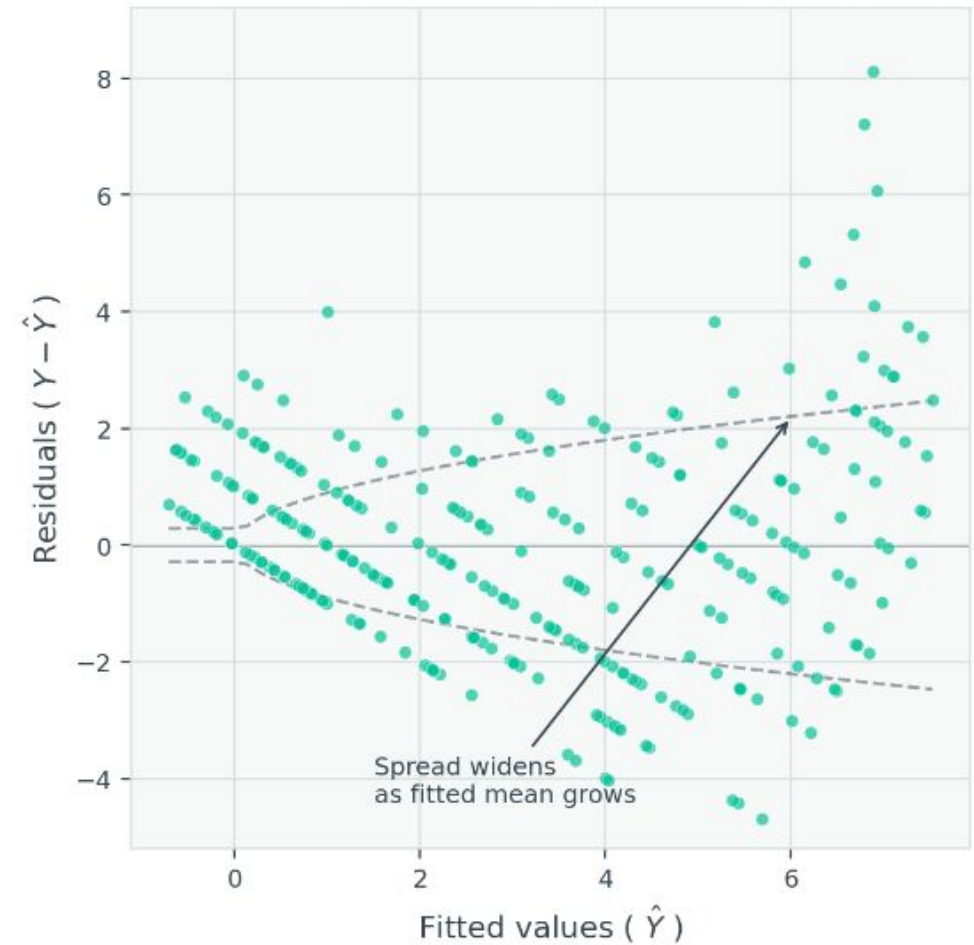
- Predictions can fall outside the physically possible range
- Estimates become inefficient or biased
- Standard errors are wrong → misleading p-values and confidence intervals
- Residual plots show clear red flags (fanning, banding)

A diagnostic thought experiment

Binary Outcome: Two Distinct Bands



Count Outcome: Fan-Shaped Spread



What we actually need

- A function linking the **mean** of Y to the predictors — not necessarily the identity/linear link
- A distribution for Y that matches how it was actually measured (binary $\rightarrow$ Bernoulli, counts $\rightarrow$ Poisson, proportions $\rightarrow$ Binomial)
- A way to let variance depend on the mean, rather than assuming it's constant

Introducing the Generalized Linear Model

Three ingredients, previewed:

1. Random component — the distribution of Y
2. Systematic component — the linear predictor
3. Link function — connects the two

Key reframe: this isn't abandoning linear regression — it's generalizing it.

1. Random component — the probability distribution of Y (from the exponential family):

- Binary outcome $\rightarrow$ Binomial
- Count outcome $\rightarrow$ Poisson
- Continuous positive $\rightarrow$ Gamma, Inverse Gaussian

2. Systematic component — the linear predictor:

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

3. Link function $g(\cdot)$ — connects the mean $\mu = E(Y)$ to the linear predictor: $g(\mu) = \eta \Leftrightarrow \mu = g^{-1}(\eta)$

Familiar distributions as members of one family

Distribution	Canonical parameter θ	Canonical link $g(\mu)$	Typical use
Normal	μ	identity: μ	continuous, unbounded
Bernoulli/Binomial	$\log(\mu/(1-\mu))$	logit	binary, proportions
Poisson	$\log(\mu)$	log	counts
Gamma	$-1/\mu$	inverse (or log)	positive, right-skewed

How GLMs are estimated

- No closed-form solution like OLS, except in the Normal case
- Estimated by **maximum likelihood** — find the β that makes the observed data most probable
- Software solves this iteratively (IRLS) — you won't do this by hand, but should know what's happening

Recap and what's next

- Match the model to the outcome, not the other way around
 - GLM = distribution + linear predictor + link function, fit by maximum likelihood
- **Next session:** Logistic regression — Theory and fitting in R